AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2013

Computer Science Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

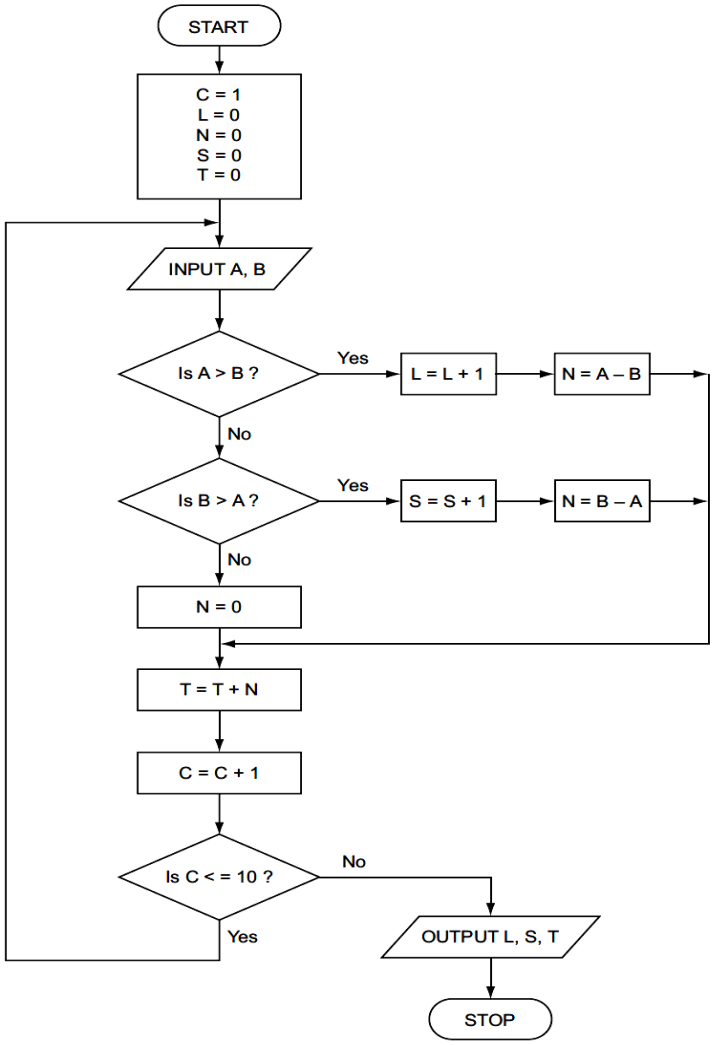
3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

Study the following flowchart very carefully.



Complete the trace table for the following data.

8,4, 3,1, 5,8, 4,2, 1,3, 2,2, 1,2, 5,5, 4,0, 5,4

C	L	N	S	T	A	B
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

Q.2. (Total 5 Marks)

State the function of the following statements of BASIC.

Statement	Function
Renum	
Left$	
Time$	
On Error Goto	
Locate (10,10)	

Q.3. (Total 5 Marks)

a. Consider the given program. (3 Marks)

```
10 INPUT “ENTER THE NUMBER ”;N
20 FOR I=1 TO 12
30      ?n;”*”;I;”=”;N*I
40 NEXT I
50 END
```

Write down the purpose of the program mentioned above.

_____

_____

_____

_____

_____

_____

b. How can the above program be modified if you want to terminate it when the value variable “I” would become 5? (2 Marks)

Line Number: _____

_____

Change Needed: _____

_____

Q.4. (Total 4 Marks)

Write a program to read the following temperature data.

35, 25, 20, 40, 7

Store values in an array called temperature and calculate their average.

Q.5. (Total 3 Marks)

What are the differences between user-defined functions and built-in functions? Describe any THREE.

	User-defined Functions	Built-in Functions
1		
2		
3		

Q.6. (Total 3 Marks)

Construct a truth table for the following Boolean expression.

$X = \overline{A}\overline{B}C * \overline{A}B\overline{C}$

A	B	C	X
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	

Q.7. (Total 8 Marks)

EITHER

a. Describe any EIGHT hardware components required for various internet applications.

OR

b. Describe a computer virus and also highlight the ways in which it affects data.

Q.8. (Total 7 Marks)

EITHER

a. Write a program that will input 25 marks and output the number of DISTINCTION, MERIT, PASS or FAIL grades.
A mark greater than 69 will get a DISTINCTION, a mark between 69 and 60 (inclusive) will get a MERIT and a mark between 59 and 50 (inclusive) will get a PASS.

OR

b. A small airport handles 400 flights per day from three airlines.
FASTAIR (code FA)
SWIFTJET (code SJ)
KNIGHTAIR (code KA)

Each flight is identified by the airline code and 3 digits. For example, FA 156. Write an algorithm, using pseudocode or otherwise, which monitors the 400 flights in and out of the airport each day. The following inputs, processing and outputs are all part of the monitoring process:

- Input flight identification.
- Calculate number of flights per day for each of the three airlines.
- Output the percentage of the total flights per day by each airline.
- Any validation checks must be included.

END OF PAPER